Bayesian Time Series Approach and Its Application to Retrieve Ground and Vegetation Variables From L-Band Passive Microwave Remote Sensing

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***Abstract*— Microwave remote sensing is a widely used and effective method for observing and understanding various land, ocean, and atmospheric processes. Information on the geophysical variables associated with these processes is typically retrieved through inversion of forward models, which describe the remotely sensed observations as functions of the geophysical variables of the scene. Inversion of the forward model is often an ill-posed problem due to the limited information content of microwave measurements and the complexity of the observed scene. The Bayesian approach provides means to incorporate prior information to reduce the ill-posedness of the problem. Considering temporal information is particularly useful in this context, as the temporal characteristics of geophysical variables can be used as prior information to reduce the ill-posedness. Furthermore, considering the temporal domain is natural due to the sequential nature of remote sensing observations. To incorporate such prior information, we introduce a Bayesian inversion of a time series of geophysical variables from a time series of remote sensing observations. The method is formulated in a general form and is therefore applicable to different remote sensing problems. To demonstrate, we applied the method to Soil Moisture and Ocean Salinity (SMOS) L-band brightness temperature measurements to simultaneously retrieve, for the first time, ground permittivity, surface roughness, vegetation optical depth, and scattering albedo over a one-year period at a northern boreal forest site. We compared the retrieved geophysical variables with the ground reference: retrieved and measured ground permittivity agreed with a correlation of 0.91, and retrieved and measured vegetation optical depth agreed with a correlation of 0.68 and the standard deviation of the difference of 0.11. We demonstrated that using the time series method, we could retrieve ground surface roughness and vegetation scattering albedo. Although reference measurements were not available for these variables, the retrieved values were consistent with the ground permittivity and vegetation optical depth. In general, ground surface roughness and vegetation scattering albedo are not retrieved in the nominal case, where data from different satellite overpasses are treated separately. This demonstrates that the time series method enables the retrieval of additional information compared to the usual approach.**

***Index Terms*— Bayes method, inverse problem, L-band, passive microwave remote sensing, soil, vegetation.**

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I. Introduction

MICROWAVE remote sensing is a widely used and successful method for studying a wide range of Earth processes [1]. The benefits of microwave remote sensing include obtaining information from remote locations where direct observations are not available, the temporal regularity and history of the observations, and the capacity to observe independently of lighting and weather conditions. However, to obtain information on the geophysical variables from an indirect observation forms its own challenge. Methods for retrieving geophysical variables from microwave observations are usually based on an inversion of empirical, semi-empirical, or physical forward models that describe the observation as a function of the geophysical variables. The microwave instruments observe upwelling radiation (passive microwave remote sensing) or the phase, amplitude, or travel time of the backscattered signal (active microwave remote sensing) [2], [3], [4].

In general, the satellite retrieval problem consists of three variables: the satellite observation y, the unknown geophysical variables x, and the known background variables b. The variables are connected through the forward model

$$y = f(x, b, \gamma) + u \tag{1}$$

where the terms u and γ represent the observation noise and the measurement configuration of the observation y, respectively. For example, in passive microwave remote sensing, y contains brightness temperatures in different observation configurations, such as the frequency, polarization, and observation angle, which are expressed in γ. The usual approach to retrieve x is to find the value minimizing the cost function, usually defined as

$$\text{cost}(x) = \|y - f(x, b, \gamma)\|_u^2 + \alpha(x) \tag{2}$$

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where $\|\cdot\|_u$ denotes a norm of the observation-model mismatch weighted by the observation uncertainty u. The term $\alpha(x)$ is referred to as the guiding term, the regularization term [5], or the prior term [6], [7]. The role of the term $\alpha(x)$ is to constrain the retrieval based on prior information or belief about the unknown geophysical variables in x.

A common challenge in microwave remote sensing is the limited information content of the observations relative to the complexity of the observed scene. Often, there are more unknown geophysical variables than can be reliably retrieved from the data. This information deficit can be mitigated by incorporating appropriate prior information about the unknown variables. For example, [8] illustrates this "need for prior" in the context of passive microwave remote sensing of snow. The Bayesian framework enables the transparent integration of prior information into the retrieval process. Moreover, it provides a natural means to quantify uncertainties in both the measurements and the retrieved solution. A key advantage of the Bayesian framework is that both the prior information and the uncertainties have a clear interpretation in terms of probability distributions.

A typical approach is to perform the retrieval in a one-by-one manner for each satellite overpass separately (see [2], [3]), as it is expressed in (1) and (2). However, this limits the type of prior information that can be included in the retrieval process through $\alpha(x)$ in (2). To incorporate information about the temporal characteristics of the geophysical variables, we considered a retrieval problem where a time series of observations y_k from satellite overpasses at times t_k were considered jointly. This allows, for example, to incorporate information about potentially time-invariant variables, which significantly reduces the number of unknowns.

A time series approach has been previously investigated in [9], [10], and [11], where the authors considered the retrieval of ground and vegetation variables from multiple satellite overpass L-band brightness temperature observations. Their approach was based on a specific assumption about the temporal characteristics of the retrieved ground and vegetation variables. Ebtehaj and Bras [12] considered a similar retrieval problem and regularization imposed on the temporal behavior of the geophysical variable. Durand et al. [13] used a sequential time series approach to incorporate prior information about the temporal characteristics of the snow water equivalent and snow microstructure to improve the retrieval of the former.

The primary focus of this study is to present a time series retrieval method from a more general mathematical perspective, not specific to particular geophysical variables or observations. The method is formalized in the Bayesian framework [6], [7]. The motivation of this study originates from the following practical question: can we retrieve additional information by considering a time series of observations simultaneously, compared to considering them independently? The use of the method is demonstrated by applying it to a time series of L-band brightness temperature observations from the European Space Agency's Soil Moisture and Ocean Salinity (SMOS) satellite [14] to retrieve soil and vegetation variables. Traditionally, it is considered possible to retrieve ground permittivity (related to soil texture, moisture, and freeze-thaw state) and vegetation optical depth (related to vegetation biomass, water content, and freeze-thaw state) from brightness temperatures measured during a single SMOS overpass, given values of two time-invariant variables; ground surface roughness and vegetation single scattering albedo [3], [11], [15].

Although current retrieval algorithms are relatively mature, and the soil moisture retrieval performance of the SMOS and NASA's Soil Moisture Active Passive (SMAP) [16] satellites is generally considered satisfactory in nonforested areas [17], [18], several challenges remain. These include mitigating temporally dependent errors [19], addressing crosstalk between soil moisture and vegetation [20], and accounting for the effects of forest vegetation and organic soils [21], [22]. A key issue is the reliance on ancillary data sources for ground surface roughness and vegetation single scattering albedo, which often fail to accurately represent L-band emission characteristics.

This study aims to show that with the introduced time series approach, it is possible to retrieve additional information, namely, the ground surface roughness and vegetation single scattering albedo, avoiding the typical assumptions used in contemporary L-band retrieval algorithnms to define these parameters (see [3], [23]). The retrieved ground permittivity and vegetation optical depth are compared with ground reference measurements made at the Sodankylä Arctic Space Center of the Finnish Meteorological Institute. For an overview on the passive microwave remote sensing of soil and vegetation variables, we refer to [1] and [24].

II. Time Series Retrieval in The Bayesian Framework

A. Bayesian Formulation

In the following, we follow the Bayesian framework in defining the retrieval problem [6], [7]. Let the vectors $y_k \in \mathbb{R}^{m_k}$ denote the satellite measurements at the observation times t_k, $k = 1, \ldots, K$. The geophysical variables of interest and the known background variables at time t_k are denoted by $x_k \in \mathbb{R}^n$ and $b_k \in \mathbb{R}^\ell$, respectively. The geophysical time series space $X \subset \mathbb{R}^N$ consists of the time series $x = (x_1, \ldots, x_K)$ and therefore has dimension $N = Kn$. The geophysical time series is parameterised with a transformation function $x = \varphi(w)$, where $\varphi : \mathbb{R}^s \longrightarrow X$ is a mapping between the s-dimensional retrieval space and the physical time series state space. The motivation for this parameterization is described in Section II-C. The relationships between the observation, the geophysical variables, and their parameterization are described by

$$y_k = f(x_k, b_k, \gamma_k) + u_k \tag{3}$$

$$x_k = \varphi_k(w) \tag{4}$$

where $f(\cdot)$ is the forward model, u_k is the observation noise, and γ_k contains the measurement configurations corresponding to the observation y_k. The variables u_k and w are modeled as Gaussian random variables

$$u_k \sim \mathcal{N}(0, S_k) \tag{5}$$

$$w \sim \mathcal{N}(0, I). \tag{6}$$

Assuming the stochastic indenpendence of u_k and w, (3)–(6) define the retrieval problem.

The posterior distribution of w is given by the Bayes theorem

$$p(w|y, b, \gamma) \propto p(y|w, b, \gamma)p(w). \tag{7}$$

The likelihood of the time series $y = (y_1, \ldots, y_K)$ appearing in (7) is given by

$$p(y|w, b, \gamma) = \prod_{k=1}^{K} \mathcal{G}(y_k - f_k; S_k) \tag{8}$$

where $f_k = f(\varphi_k(w), b_k, \gamma_k)$ and $\mathcal{G}(\cdot; S)$ denotes the Gaussian density function with zero mean and covariance matrix S. We solved the problem by first approximating the posterior distribution of w, given by (7), and then mapping the approximate posterior distribution of w back to the geophysical variable space with $x = \varphi(w)$.

We approximated the posterior distribution of w as a Gaussian random variable, with its mean given by the maximum a posteriori estimate of w, given by the minimum of the cost function, which is defined as the negative log-posterior density

$$\text{cost}(w) = -\log p(w|y, b, \gamma). \tag{9}$$

Using (6)–(8), this becomes

$$\text{cost}(w) = \frac{1}{2}\sum_{k=1}^{K} \|y_k - f(\varphi_k(w), b_k, \gamma_k)\|_{S_k}^2 + \frac{1}{2}\|w\|^2 \tag{10}$$

where $\|r\|_{S_k}^2 = r^\top S_k^{-1} r$. Due to the nonlinearity of the forward model $f(\cdot)$, minimizing the cost function is a nonlinear least squares minimization problem. This is solved by an iterative algorithm, discussed in Section II-B.

We approximated the posterior covariance $S_{w|y}$ of w with the Laplace approximation (see [25])

$$S_{w|y}^{-1} = D_w^2 \text{cost}(w) \tag{11}$$

where D_w^2 denotes the Hessian with respect to w. The dimensionality of the problem can make computing the Hessian in (11) using finite differences computationally prohibitive. See the Appendix for computing the Hessian in an efficient manner, as well as other relevant computational aspects.

In our approach, we fixed the prior distribution of w and chose φ to obtain the desired prior distribution for the geophysical time series x. The distributions are related via the change of variable formula for probability densities

$$q(x) = p\big(\varphi^{-1}(x)\big) D\varphi^{-1}(x) \tag{12}$$

where $q(\cdot)$ and $p(\cdot)$ are the posterior density functions of x and w, respectively. In the Bayesian worldview, the posterior distribution of x is considered to be the solution to the retrieval problem. To achieve this, (12) can be used with the Gaussian approximation of the posterior density of w. For practical solutions, one can either use the linearization of the transformation function $\varphi(\cdot)$ around the $w_{\min}$, or alternatively use a Monte Carlo approach to draw samples of w from its approximate posterior distribution $\mathcal{N}(w_{\min}, S_{w|y})$ and map them to samples of $x = \varphi(w)$. With certain special transformation functions $\varphi(\cdot)$, direct formulas for the posterior of x are available. In the example presented in this article, this is the case, and the other options are not discussed further.

B. Iterative Solution and Computational Aspects

We seek to find the maximum a posteriori estimate of w by finding the minimum of the cost function defined in (10). As before, we denote the time series of observations by $y = (y_1, \ldots, y_K)$, the corresponding modeled time series by $f = (f_1, \ldots, f_K)$, the observation errors by $u = (u_1, \ldots, u_K)$, and the covariance matrix of u by S_u. By defining the observation part of the scaled residuals function $F_y(w) = S_u^{-1/2}(y - f) \in \mathbb{R}^M$, where $S_u^{-1/2}$ is such that $(S_u^{-1/2})^\top S_u^{-1/2} = S_u^{-1}$, then

$$F(w) = \begin{bmatrix} F_y(w) \\ w \end{bmatrix} \tag{13}$$

defines the total scaled residuals function. The cost function defined in (10) can now be written as

$$\text{cost}(w) = \frac{1}{2}\|F(w)\|^2. \tag{14}$$

If the noise in the measurement channels are independent (i.e., the components of u_k for all k), then $S_u^{-1/2} = \text{diag}(1/\sigma_1, 1/\sigma_2, \ldots, 1/\sigma_M)$ where σ_i are the standard deviations of the components of u.

Because the cost function can be written as a sum of squares given by (14), we can use nonlinear least squares optimization algorithms to find the solution $w_{\min}$. We used the Levenberg–Marquardt algorithm [26], [27] with the `lsqnonlin.m` implementation from MATLABs optimization toolbox [28], [29]. The Levenberg–Marquardt algorithm requires the derivative matrix $D_w F(w)$ at each iteration. Computing this by naive finite difference approximation is computationally expensive, and unfeasible in our case. However, by exploiting the fact that in any satellite remote sensing data, the forward model response at time t to a change in the geophysical variables at different times t' is zero, the derivative $D_w F(w)$ can be computed considerably faster. For technical details, see the Appendix.

C. Role of the Transformation Function

The transformation function $x = \varphi(w)$ between the retrieval space $\mathbb{R}^s$ and the geophysical time series space $X \subset \mathbb{R}^N$ was introduced in the retrieval problem. The transformation function plays several roles. First, it allows us to consider a lower dimensional subspace of the time series space X. For example, some of the geophysical variables can be considered time-invariant (but still possibly variable in space), thus reducing the degrees of freedom of the retrieved variables. Furthermore, considering the temporal correlation of the geophysical variables can be used to reduce the dimensionality of the problem, thus reducing the computational burden.

Second, for computational purposes, we are motivated to use a Gaussian prior distribution for the retrieved variables. However, the geophysical variable x might not be expected to follow such a distribution. Retrieving w with a convenient transformation $\varphi(\cdot)$ can help to produce a suitable prior

distribution for x, while still benefiting from the Gaussian prior for w. Furthermore, the physical constraints on x do not need to be considered in the iterative algorithm.

Third, the Gaussian approximation of the posterior distribution of x may not be appropriate for the same reasons given above. With a convenient choice of $\varphi(\cdot)$, approximating the posterior of w as a Gaussian distribution and transforming this to x results in a more suitable distribution. It should be noted that while the expected value of the posterior distribution of x is invariant under coordinate transformations (under some general conditions), the maximum a posteriori estimate is not, unless the transformation function is affine. Finally, it normalizes the retrieved variable w, potentially improving the convergence of the iterative algorithm.

III. Application to SMOS Data

To demonstrate the introduced time series retrieval method, we applied it to a specific land remote sensing retrieval problem. Our goal was to retrieve time series of four geophysical variables: ground permittivity ε_{g}, ground surface roughness h_{g}, vegetation optical depth τ_{v}, and vegetation single scattering albedo ω_{v} from a time series of L-band brightness temperature measurements. To apply the approach, we define the items appearing in (3)–(6), describing the retrieval problem. In the notation of these equations, we have

$$x_k = \begin{bmatrix} \varepsilon_{\mathrm{g}}(t_k) \\ \tau_{\mathrm{v}}(t_k) \\ h_{\mathrm{g}}(t_k) \\ \omega_{\mathrm{v}}(t_k) \end{bmatrix}, \quad y_k = \begin{bmatrix} T_{\mathrm{B}}\left(\gamma_k^1, t_k\right) \\ T_{\mathrm{B}}\left(\gamma_k^2, t_k\right) \\ \vdots \\ T_{\mathrm{B}}\left(\gamma_k^{m_k}, t_k\right) \end{bmatrix} \tag{15}$$

where the vector $\gamma_k = \gamma(t_k)$ includes known measurement parameters, affecting the measured brightness temperature. In what follows, we describe the SMOS measurements used for $T_{\mathrm{B}}(\gamma_k, t_k)$ together with the measurement configuration variables γ_k, the forward model $f(\cdot)$, and the error terms u_k.

A. SMOS Brightness Temperatures & Forward Model

For the L-band brightness temperature observations, we used the L1C data product from the SMOS satellite [14], [30]. A time series of SMOS observations was extracted for the Sodankylä site for a one-year period, from September 1, 2021, to September 1, 2022. During a single overpass at the time t_k, the SMOS satellite measures a sequence of brightness temperatures denoted by $T_{\mathrm{B}}(\gamma_k^i, t_k)$ with different measurement configurations denoted by $\gamma_k^i = \gamma^i(t_k)$, $i = 1, \ldots, m_k$. The associated vectors are denoted by $T_{\mathrm{B}}(\gamma_k, t_k)$ and γ_k, respectively. The measurement configurations include the observation angle, the brightness temperature polarization, the elliptical footprint shape and orientation, and the total rotation angle from the ground reference frame to the satellite reference frame, due to geometric and Faraday rotations. The measurement uncertainties related to $T_{\mathrm{B}}(\gamma_k, t_k)$ are provided in the data. These are used to define the covariance matrices S_k of the noise terms u_k in (3) and (5), details of which are described in Section III-B. Fig. 1(a) shows a time series of linearly polarized brightness temperature from SMOS with observation angle between $\theta = 45° \pm 2.5°$.

To model the SMOS brightness temperature, we used the ground and snow microwave emission model [31] extended with a vegetation layer [32]. The horizontally and vertically polarized upwelling brightness temperatures at the top of the atmosphere were modeled as weighted averages of submodels representing different land cover contributions

$$f(x_k, b_k, \gamma_k) = \sum_{\text{type,snow}} \vartheta_{\text{type}}^{\text{snow}} f(x_k, b_k, \gamma_k; \text{type, snow})$$
$$\sum_{\text{type,snow}} \vartheta_{\text{type}}^{\text{snow}} = 1. \tag{16}$$

Above, the summation is over the six submodels based on land cover type: open, forest, or water, with a binary snow state: snow-covered or snow-free scene. The areal land cover fractions $\vartheta_{\text{type}}^{\text{snow}}$ were computed separately for each SMOS measurement based on the measurement footprint. The background variable vector b_k consists of the physical temperatures of the scene

$$b_k = \begin{bmatrix} T_{\mathrm{g}}(t_k) \\ T_{\mathrm{v}}(t_k) \\ T_{\mathrm{a}}(t_k) \\ T_{\mathrm{w}}(t_k) \end{bmatrix} \tag{17}$$

where the subscripts refer to (g)round, (v)egetation, (a)ir, and (w)ater. In addition, the snow was modeled assuming a dry snow pack with a constant density of $\rho_s = 250$ kg/m^3. This represents a typical density value in the region of interest. In addition, the error caused by variable snow density is minor [33], [34]. Snow wetness was omitted in this study for two reasons. First, modeling its contribution to the brightness temperature reliably is challenging due to the lack of appropriate in situ data and the complexity of the snowpack under melting conditions. Second, retrieving snow wetness is similarly challenging for the same reason; the complexity of the snowpack during melt.

The areal land cover fractions $\vartheta_{\text{type}}^{\text{snow}}$ in (16) were computed based on the orientation ϕ and the semi-major and minor axes a and b of the elliptic SMOS footprint. These were provided in the SMOS L1C data, and included in γ. The footprints were modeled as 2-D Gaussians with the mean equal to the footprint center and the covariance matrix constructed with

$$S_{\text{footprint}} = V D V^{\top}$$
$$V = \begin{bmatrix} \sin(\phi) & \cos(\phi) \\ \cos(\phi) & -\sin(\phi) \end{bmatrix}, \quad D = \begin{bmatrix} a^2 & 0 \\ 0 & b^2 \end{bmatrix}. \tag{18}$$

The radiometric land cover fractions $\vartheta_{\text{type}}^{\text{snow}}$ were estimated by integrating the Gaussian footprint pattern over the high-resolution land cover maps. Fig. 2 shows averaged observation footprints from a single SMOS overpass over the Sodankylä region.

Finally, brightness temperatures modeled in the ground reference frame were transformed to the SMOS satellite reference frame with [35]

$$\begin{bmatrix} T_{\mathrm{B,X}} \\ T_{\mathrm{B,Y}} \end{bmatrix} = \begin{bmatrix} \cos^2(\alpha) & \sin^2(\alpha) \\ \sin^2(\alpha) & \cos^2(\alpha) \end{bmatrix} \begin{bmatrix} T_{\mathrm{B,H}} \\ T_{\mathrm{B,V}} \end{bmatrix}. \tag{19}$$

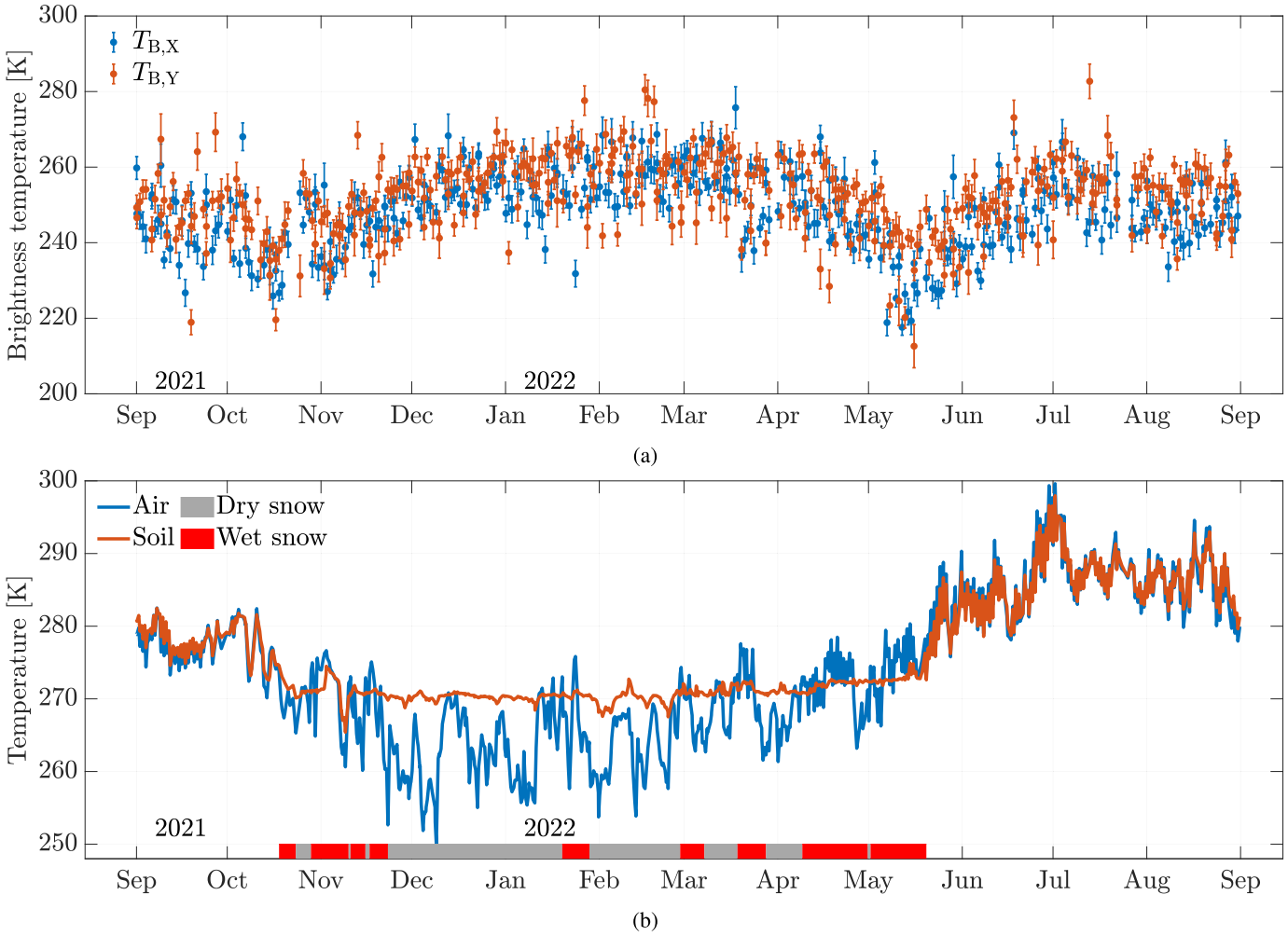


Fig. 1. (a) Time series of SMOS brightness temperatures for the one-year period from the observation angles $\theta = 45° \pm 2.5°$ over the Sodankylä region (see Fig. 2). The error bars show the radiometric accuracies of the measurements (used as the uncertainty). (b) Time series of ERA5 Land soil and air temperatures and IMS-derived snow flags for dry and wet snow.

The rotation angles α were included in γ. This was applied separately to each modeled brightness temperature in the profile measured by SMOS during one overpass. Depending on the polarization of that measurement (also included in γ), either $T_{B,X}$ or $T_{B,Y}$ was used for that component of $f(x_k, b_k, \gamma_k)$. The resulting modeled brightness temperature profile $f_k = f(x_k, b_k, \gamma_k)$ corresponds to the measured profile $y_k = T_B(\gamma_k, t_k)$ in (15). For details on the implementation of the forward model, see [34].

B. Time Series Retrieval

We used the following permutation of the geophysical variables in the time series x for ease of presentation and implementation:

$$x = \begin{bmatrix} \varepsilon_g \\ \tau_v \\ h_g \\ \omega_v \end{bmatrix}. \tag{20}$$

The first K components of x form the ground permittivity time series, the next K components form the vegetation optical depth time series, and so forth.

We assumed the uncertainties provided in the SMOS data were independent Gaussian errors and defined the covariance matrices S_k of the noise terms u_k as diagonal matrices with entries from the data together with an additional additive uncertainty term to account for modeling errors in the forward model and image reconstruction errors in the processing of the L1C data. By examining the distribution of the scaled model-observation residuals (not shown), we found the additive Gaussian uncertainty with a standard deviation of $\sigma_{add} = 4$ K to be an appropriate choice.

The transformation function $\varphi(\cdot)$ was constructed to incorporate prior information about the geophysical variables into the retrieval problem. We made the following a priori assumptions regarding the retrieved geophysical variables.

1) Ground permittivity and vegetation optical depth are continuous in time with a limited rate of change.
2) Ground surface roughness and vegetation scattering albedo are time-invariant variables.
3) The four geophysical variables are stochastically independent of one another.

In addition, have the following physical fact.

4) All geophysical variables are positive.

The above a priori information was encoded using the transformation function

$$\varphi(w) = \exp(Aw + m) \tag{21}$$

where

$$m = \begin{bmatrix} m_\varepsilon \mathbf{1} \\ m_\tau \mathbf{1} \\ m_h \mathbf{1} \\ m_\omega \mathbf{1}. \end{bmatrix}, \quad A = \begin{bmatrix} A_\varepsilon & 0 & 0 & 0 \\ 0 & A_\tau & 0 & 0 \\ 0 & 0 & A_h & 0 \\ 0 & 0 & 0 & A_\omega. \end{bmatrix}. \tag{22}$$

The A_ε and A_τ are defined as the factorizations of the following covariance matrices:

$$A_\varepsilon A_\varepsilon^\top = \mathrm{cov}\big(\log \varepsilon_\mathrm{g}\big), \;\; A_\tau A_\tau^\top = \mathrm{cov}\big(\log \tau_\mathrm{v}\big).$$

These covariance matrices were modeled by

$$\begin{aligned} \mathrm{cov}\big(\log \varepsilon_\mathrm{g}\big)_{i,j} &= \mathrm{cov}\big(\varepsilon_\mathrm{g}(t_i), \varepsilon_\mathrm{g}(t_j)\big) \\ &= \sigma_\varepsilon^2 \exp\left(-\frac{|t_i - t_j|}{\lambda_\varepsilon}\right) \end{aligned} \tag{23}$$

and the covariance matrix $\mathrm{cov}(\log \tau_\mathrm{v})$ was defined similarly. The submatrices A_h and A_ω were defined as

$$A_h = \sigma_h \mathbf{1}, \quad A_\omega = \sigma_\omega \mathbf{1} \tag{24}$$

where $\mathbf{1} \in \mathbb{R}^{K\times 1}$ with all entries equal to one. The parameters m, σ, and λ describe the transformation function φ, and through that, the prior distribution of $x = \varphi(w)$. The values used in this study are shown in Table I.

We chose the form in (21) in particular because it reflects the positivity constraint on the geophysical variables. In this case, the Gaussian prior for w corresponds to a log-Gaussian prior for x. Also, by approximating the posterior of w as a Gaussian distribution through (11), we obtained a log-Gaussian approximation of the posterior distribution of x. The mode, mean, median, and probability intervals of the posterior x all have closed-form expressions in terms of the posterior mode w_min and approximated covariance matrix $S_{w|y}$. We used the median and 95% probability interval of the approximated marginal posterior distribution; these are given by

$$\begin{aligned} x_\mathrm{median} &= \exp(\mu) \\ x_\mathrm{low} &= \exp\big(\mu + \zeta\, \Phi^{-1}(q)\big) \\ x_\mathrm{high} &= \exp\big(\mu + \zeta\, \Phi^{-1}(1-q)\big) \end{aligned} \tag{25}$$

where $q = 0.025$, $\mu = A w_\mathrm{map} + m$, $\zeta^2 = \mathrm{diag}(A S_{w|y} A^\top)$, and $\Phi(\cdot)$ is the standard Gaussian distribution function.

The prior parameters m and σ describe the prior mean and standard deviation of the geophysical time series, and λ describes the prior temporal characteristics of the variable, i.e., what is its rate of change. While m and σ are in logarithmic units of the corresponding geophysical variable, λ parameters are in time units (measured here in days).

Finally, we used only part of the available SMOS data with the following criteria: first, we used only the observations from the snapshots that measure only linear polarization. Second, we used measurements from ascending orbits. This restriction was made because the descending orbit is more susceptible to radio frequency interference over the Sodankylä region [36]. Furthermore, using a single orbit removes the effects of diurnal variations from the signal, making the used priors more appropriate. Third, the measurements that were flagged as susceptible to radio frequency interference in the L1C data were removed. Last, for each overpass, we removed potential outliers from the brightness temperature profile by fitting a second-order polynomial to the brightness temperature as a function of the observation angle and removing instances where the brightness temperature value was more than five observation error standard deviations (provided in the data) away from the polynomial. In total, the time series considered in this study contained data from $K = 413$ SMOS overpasses with the total number of measurements $M = 19\,469$.

TABLE I

PRIOR DISTRIBUTION PARAMETERS FOR THE GROUND PERMITTIVITY ε_g AND SURFACE ROUGHNESS h_g, VEGETATION OPTICAL DEPTH τ_v, AND SCATTERING ALBEDO ω_v. THE SUBSCRIPTS g AND v ARE OMITTED FOR SIMPLICITY

Parameter	Value	Parameter	Value
m_ε	$\log(10)$	m_h	$\log(0.3)$
σ_ε	0.2	σ_h	0.5
λ_ε	3		
m_τ	$\log(0.4)$	m_ω	$\log(0.06)$
σ_τ	0.2	σ_ω	0.5
λ_τ	1		

IV. OTHER DATA

A. Physical Temperatures

The physical temperatures of the observed scene are the sources of the measured microwave radiation, and were considered as known background variables in this study. We used the air temperature at 2 m above ground and ground temperature at 0–7 cm depth from the ECMWF data product [37]. Data were provided on a Gaussian N400 grid at 6 h intervals. Data were interpolated to the satellite observation grid node and overpass time using bi-linear and linear interpolation, respectively. For the forward model, the temperatures were included in b_k in (17). We assumed that the vegetation temperature is equal to the air temperature $T_\mathrm{v} = T_\mathrm{a}$ and the water temperature is equal to the ground temperature $T_\mathrm{w} = T_\mathrm{g}$. Fig. 1(b) shows time series of the physical temperatures used in this study.

B. Land Cover Information

Modeling the observed brightness temperature requires knowledge of the land cover distribution within the observed scene. As described earlier, six different submodels were used for open, forested, and water scenes, each with and without snow. Two datasets were used to compute the areal land cover fractions associated with each of the submodels: the Corine land cover dataset from the Finnish Environment Institute [38], and the Interactive Multisensor Snow and Ice Mapping System (IMS) daily northern hemisphere snow and ice analysis product from U.S. National Ice Center [39].

The Corine land cover data are provided in the Universal Transverse Mercator projection at a 20 m resolution. The land cover classes were merged into four categories: open, wetland, forest, and water. In addition, the wetland class was considered

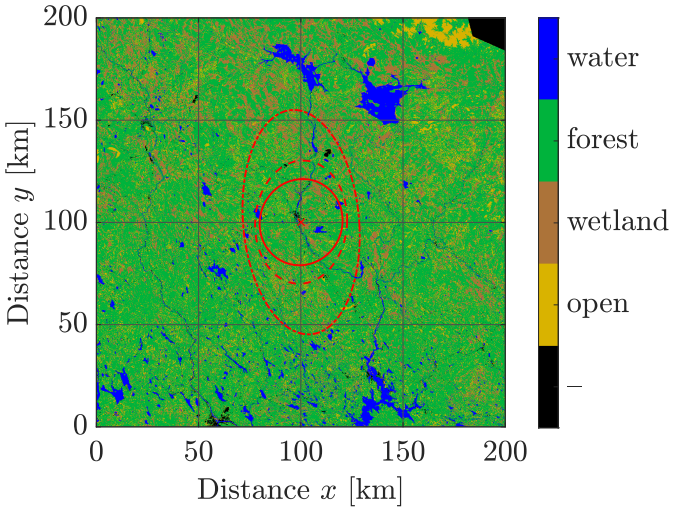


Fig. 2. 200 × 200 km working area with reclassified Corine land cover distribution and SMOS footprints from different observation angles during an overpass of the Sodankylä region. The red ellipses represent footprints corresponding to different observation angles during the same overpass: from innermost to outermost, they correspond to angle ranges of 25°–30°, 40°–45°, and 55°–60°.

as open in the forward modeling. The presence of snow cover was determined from the IMS snow product. The data are provided in daily temporal and 4 × 4 km spatial resolution.

The radiometric fraction of each land cover class was determined by integrating the SMOS footprint patterns over the land cover map. Fig. 2 shows the working area with the reclassified land cover map and examples of SMOS footprints from different observation angles of a single overpass over the Sodankylä region.

We omitted wet snow in the forward model; that is, we assumed a dry snowpack when snow was present, thus introducing errors in the retrievals under wet snow conditions. To flag such cases in our analysis of the results, we used the ERA5 air temperature and IMS snow cover data. The scene was considered to be contaminated by wet snow if more than 50% of the scene was covered by snow and the air temperature had exceeded 273 K in the previous 48 h.

C. Reference Data

We compared the retrieved geophysical variables with in situ measurements from the Sodankylä Arctic Space Center of the Finnish Meteorological Institute. For ground permittivity, we used measurements from automated soil moisture network stations located around the Sodankylä region [40]. Within the study period, data were available from 10 soil moisture stations, each with 2 to 5 sensors placed 5 cm below the soil surface; 27 in total. The sensors measured at different frequencies within 50–200 MHz range. The sensors at each station were averaged to obtain a single time series for each station.

The ground permittivity time series measured at different stations can show distinctly different behavior due to the different soil, land cover, and weather conditions at the sites of the in situ stations. For this reason, and to make an efficient comparison between the retrieved and in situ ground permittivities, we computed a weighted average time series of the in situ measurements

$$\varepsilon_{\mathrm{g}}(t) = \sum_{i=1}^{4} a_i \varepsilon_{\mathrm{g},i}(t) \tag{26}$$

where the sum is over the four soil types present in the region, excluding wetlands. We used the weights a_i derived in [34, (Table B.9)] for the Sodankylä region. These weights represent the areal fraction of each soil type within the region.

As a reference for the retrieved vegetation optical depth, we used vegetation optical depth derived from local ground-based L-band radiometer observations [41], [42]. The vegetation optical depth of the forest canopy was derived following the simple radiative transfer approach outlined in [43]. While the ground-based radiometer measured horizontal and vertical polarizations from multiple observation angles, we decided to use only data from observation angle $\theta = 130°$ (with respect to nadir) and vertical polarization for consistency. The derived values represent the optical depth of the forest canopy, excluding other vegetation contributions and neglecting scattering effects within the canopy. Although the values represent a single location within the satellite footprint, they are expected to show similar temporal dynamics with the retrieved vegetation optical depth.

Ground surface roughness and vegetation single scattering albedo measurements were not available. However, both affect the retrieved ground permittivity and vegetation optical depth. Therefore, we used the comparison of the retrieved ground permittivity and vegetation optical depth with their respective reference measurements to evaluate the consistency of the retrieved surface roughness and scattering albedo. This served as an indirect assessment of the quality of the retrievals.

V. Results and Discussions

Fig. 3 shows the time series of the retrieved ground permittivity and averaged in situ time series from soil moisture stations located in the Sodankylä region. The in situ time series are color-coded by the soil type that the station represents. The black line shows the median of the retrieved time series and the shaded area shows the 95% posterior probability intervals. The right panel shows the marginal prior distribution of the ground permittivity for a single time instance (not showing temporal correlation). The in situ station time series were weighted and averaged to produce a single in situ time series representing the Sodankylä region (see Section IV-C). Fig. 4 shows a comparison between the retrieved and weighted in situ ground permittivity. The comparison shows that the retrieved ground permittivity underestimates the in situ for low values and overestimates it for the high values. However, the retrieved time series captures similar temporal dynamics, particularly the soil freezing events, which are well reflected in the lower permittivity values.

There are several reasons for the discrepancies between the retrieved and in situ measured ground permittivities as follows.

1) The soil moisture sensors operate at different frequencies (50–200 MHz) than SMOS (1400–1427 MHz). While the real part of the permittivity of free water is essentially the same at these frequencies, the bound water

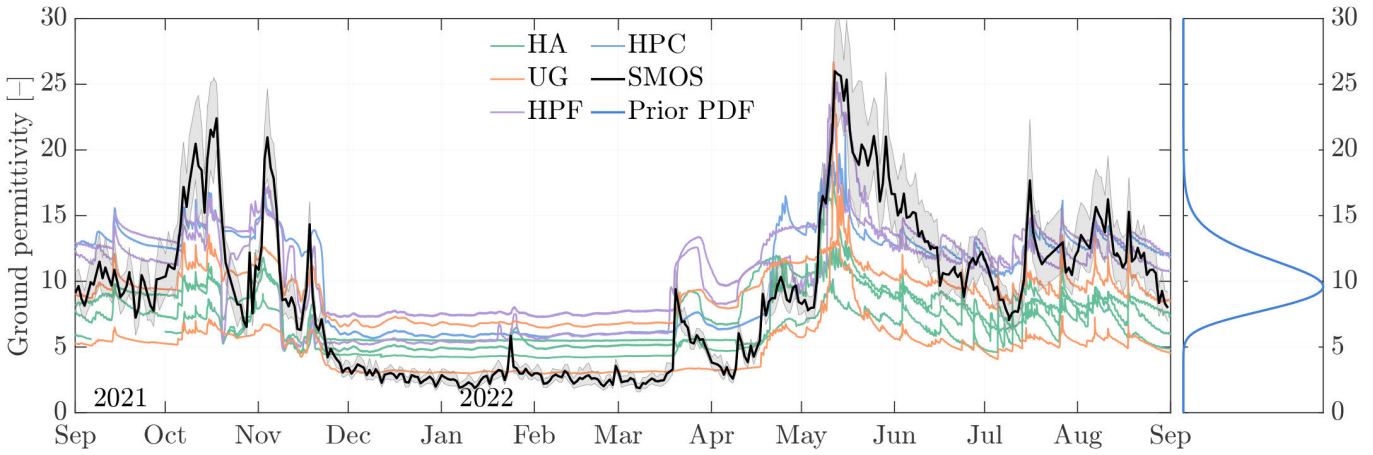


Fig. 3. Time series of the retrieved and in situ measured ground permittivity. The in situ ground permittivity time series are average values corresponding to soil moisture stations in the Sodankylä region. The in situ time series are colored based on the soil type which the station represents (HA, UG, HPF, or HPC), see [40] for more details. The right panel shows the marginal prior distribution used for the ground permittivity.

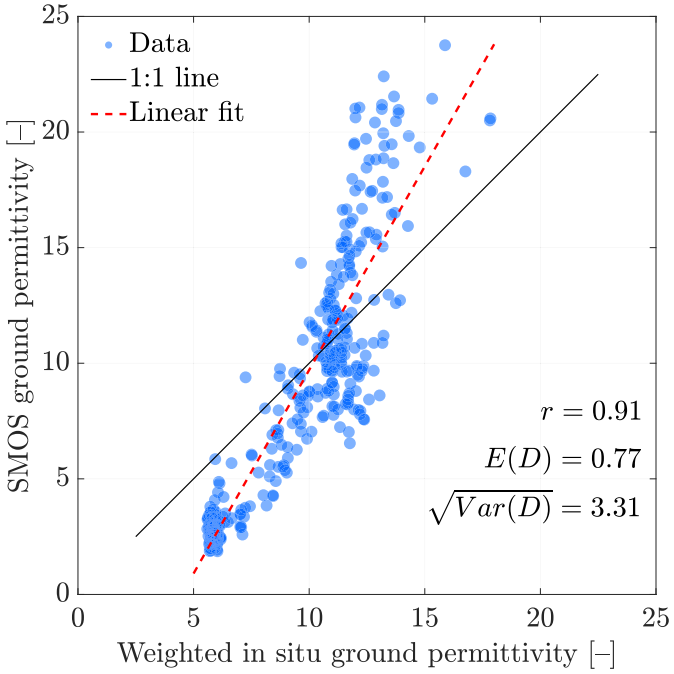


Fig. 4. Comparison between the retrieved and weighted average in situ ground permittivity. The statistics shown are the sample correlation coefficient, mean, and standard deviation of the difference.

may have frequency dependent effects, depending on the soil type and, particularly, in clayey soils [44], but less so for sandy soils [45], [46]. However, for example, [46] showed only inconclusive differences between 1.4 GHz and 50 MHz measurements for organic soils. Furthermore, characterization of the soil moisture relationship with permittivity is particularly challenging for soils with a substantial fraction of organic matter, and the models presented in the literature do not converge, not least because of the diversity of the organic soil composition [47], [48].

2) The measurements from individual sensors represent a local quantity, whereas the SMOS retrievals represent effective values over large satellite footprints. Because of the nonlinearity of the forward model, together with the assumption of spatial homogeneity of the geophysical variables used in the retrieval, the retrieved effective value may not represent the area-average estimated from the weighted in situ measurements.
3) The sensors are located 5 cm below the soil surface, where the measured permittivity may not optimally represent the retrieved L-band permittivity.
4) The wetland sites were omitted from the comparison because the sensors at 5 cm depth are often submerged in water, making them unrepresentative of the retrieved values at L-band. However, omitting them from the comparison is expected to produce its own discrepancy between the retrieved and weighted in situ ground permittivity.
5) The surface roughness parameter affects the retrieved ground permittivity. We assumed a time-invariant surface roughness, which may not hold in practice. Therefore, seasonality in the surface roughness parameter could explain some of the observed bias in the retrieved ground permittivity.

Fig. 5 shows a time series of the retrieved vegetation optical depth and that derived from local ground-based L-band radiometer measurements. The right panel shows the marginal prior distribution of the vegetation optical depth for a single time instance. The range of values is consistent between the two datasets and similar temporal dynamic can be seen. The SMOS observations are less frequent in time than the radiometer observations. This affects the comparison, especially because of the diurnal variations in the vegetation optical depth due to temperature changes [42]. The retrievals with possible wet snow conditions during the SMOS overpass (more than 50% of the scene covered by snow and air temperature exceeding 273 K within the previous 48 h) are highlighted in red. Notable discrepancies between the retrieved and ground-based vegetation optical depth are observed during the stronger melting periods.

Fig. 6 shows the same data in a scatterplot, showing only those ground-based vegetation optical depth values that are

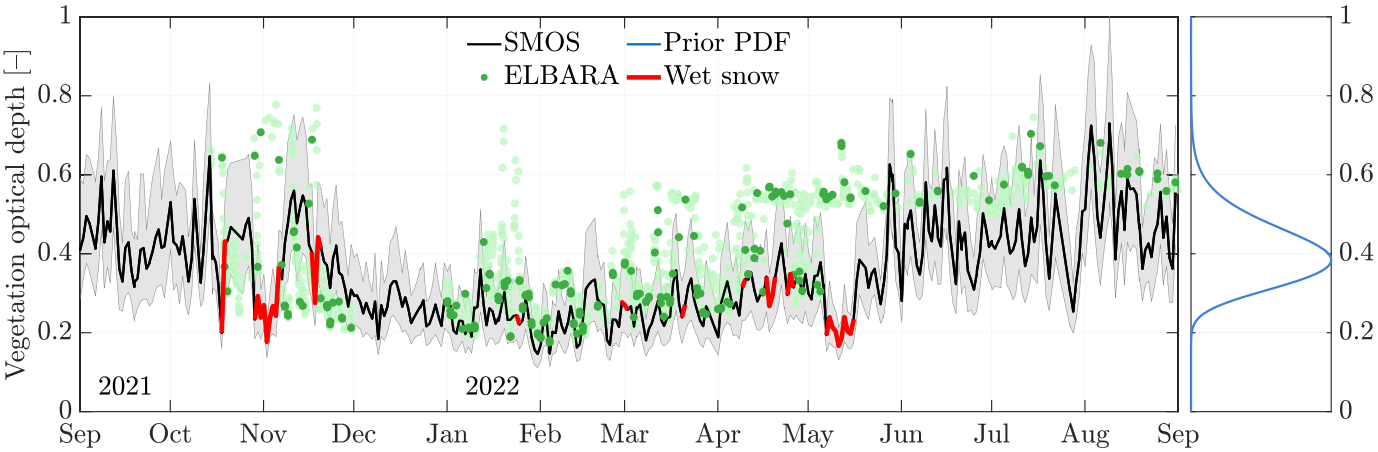


Fig. 5. Vegetation optical depth retrieved from SMOS and from the ground-based ELBARA radiometer located at the Sodankylä research site. The SMOS time series shows the posterior median (solid line) and the 95% posterior probability interval (shaded area). SMOS retrievals associated with possible wet snow conditions are highlighted in red. ELBARA-derived optical depth values measured within ±1 h of the SMOS overpass are shown in darker green. The right panel displays the marginal prior distribution used for vegetation optical depth.

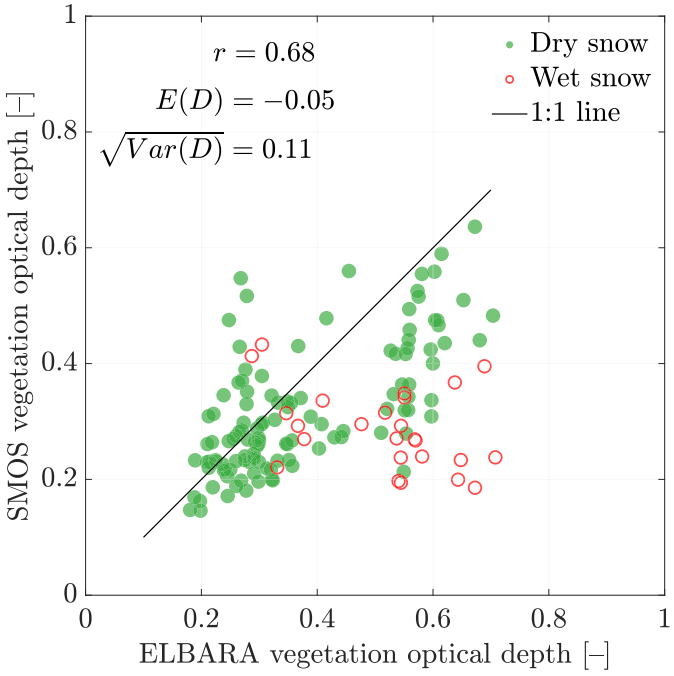


Fig. 6. Same vegetation optical depth data as in Fig. 5, plotted such that each SMOS retrieved value is compared with the temporally closest ELBARA-derived value, showing only instances where the two measurements occur within ±1 h of each other. Dry and wet snow conditions are indicated using different markers. Reported statistics; sample correlation coefficient, mean difference, and standard deviation of difference, are computed excluding the wet snow cases.

within ±1 h of the SMOS overpass, highlighting the temporal match between the SMOS vegetation optical depth and the ground-based vegetation optical depth. Since dry snow was assumed in the forward model, and liquid water in the snow has a strong influence on the observed brightness temperature, it is expected to introduce errors in the retrieved values. The retrievals associated with possible wet snow conditions are indicated in Fig. 6. When instances associated with possible wet snow were omitted, the retrieved vegetation optical depth showed good agreement with the ground-based values: mean difference −0.05, standard deviation of the difference 0.11, and correlation coefficient 0.68.

Several factors contribute to the differences between the SMOS and ground-based vegetation optical depth: First, the modeling errors in the forward modeling, e.g., spatial heterogeneity of the geophysical variables within the satellite footprint as well as the already mentioned geophysical noise from the wet snow. Second, if the SMOS overpass does not contain enough measurements, the retrieved vegetation optical depth value at that time might be more influenced by the prior than by the observation. Third, the fact that the ground-based observations are from a single location, while SMOS measures a large footprint. The temporal dynamics of the vegetation optical depth are expected to depend on the vegetation type and other properties of the forest canopy. Therefore, the dynamics might not be the same across the large footprint. Fourth, the ground-based vegetation optical depth values represent only the forest layer, whereas the SMOS retrieved values represent the total vegetation, including surface vegetation. Last, the radiometer-derived vegetation optical depth values were derived by neglecting scattering within the forest canopy [41], whereas the scattering is considered in the SMOS retrievals. On the other hand, while the scattering albedo was retrieved, we assumed it to be time-invariant. This is not likely the case as the dielectric changes of the forest canopy that cause the varying vegetation optical depth also influence the scattering properties of the forest canopy.

Reference measurements for ground surface roughness and vegetation single scattering albedo were not available. Fig. 7 shows the joint prior and the retrieved marginal posterior distributions of these two variables. The ground surface roughness prior was chosen to reflect the values used in the SMOS L2SM algorithm [3]. The scattering albedo prior was chosen with similar consideration, following the values used in the L2SM processor, but also aligned with, for example, [49]. The prior uncertainty was set high for both parameters, for scattering albedo supported by [50].

The posterior surface roughness attained a high value compared to the prior. However, the retrieved posterior was in

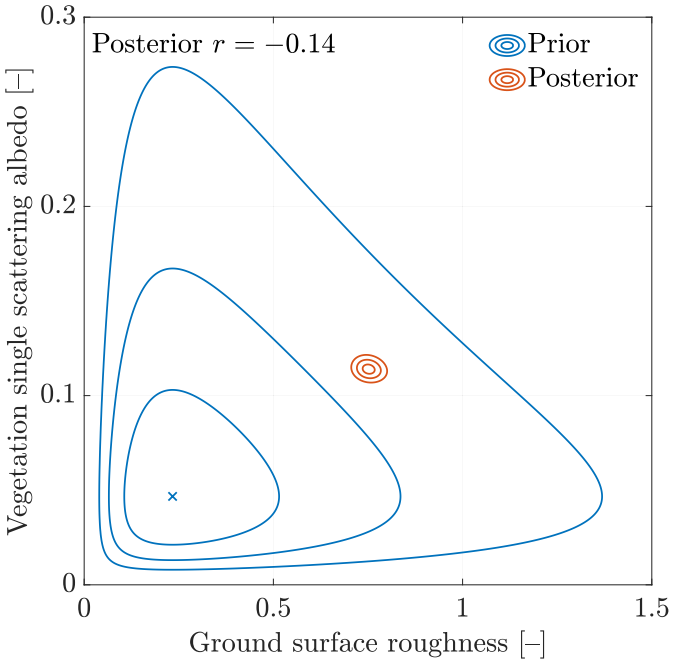


Fig. 7. Prior and approximate posterior distributions of the ground surface roughness h_g and the vegetation single scattering albedo ω_v. The three contour levels shown for each distribution correspond to the 1, 2, and 3 standard deviation curves of the respective normal distributions. The correlation coefficient of the approximated joint posterior distribution is also shown.

agreement with $h_g = 0.8$ used in [34]. Two reasons for the difference between the prior and posterior surface roughness are: First, the winter period with snow was included, which is not the usual case for SMOS soil moisture retrievals. In winter, the penetration depth of the radiation into the ground is increased due to freezing, possibly adding a so-called dielectric roughness effect reflecting small-scale dielectric heterogeneities [24]. Also, dry snow can theoretically increase the roughness effect due to its wavelength dependence: the wavelength of the radiation is reduced inside the snow, and therefore, at the snow-ground interface, increasing the roughness effect [51]. The second reason is that the forward model used in this study is based on 2-flux radiative transfer, where both up and downwelling radiation, together with multiple reflection and scattering effects of the two are considered [31], [32], [52], [53], whereas many SMOS and SMAP studies use a simpler zeroth order radiative transfer model.

Similarly, for the scattering albedo, the retrieved posterior shows higher values compared to the prior. The retrieved posterior was consistent with the values used in previous studies focusing on the Sodankylä region [34], [42] as well as theory [54]. The difference with the value used in the L2SM processor is again explained by the difference in the radiative transfer modeling, as the scattering is modeled fundamentally different in the simpler models and the more physically-based 2-flux approach [32]. The consideration of the winter period may also be a contributing factor, as the scattering albedo of a frozen forest may be different from that of the nominal case [54].

Previous studies that have estimated the scattering albedo from satellite L-band brightness temperature data include [9], [10], [49], [50]. The advantages of the method presented in this article are all rooted in the Bayesian approach: the description of the prior information and the estimation of the posterior uncertainties. Furthermore, the method presented here is general and can be applied to different microwave remote sensing problems.

Both the scattering albedo and the ground surface roughness have been calibrated with in situ data in numerous studies, for example [55]. In such approaches, the surface roughness and vegetation scattering albedo are considered as empirical tuning parameters that optimize the agreement between the retrieved and in situ measured geophysical variables. Such a calibration depends on many factors, such as the retrieved variable used in the calibration (usually only soil moisture) and different modeling errors, for example, from the dielectric mixing models that describe the dielectric constant of the soil as a function of the soil moisture. In essence, the calibration parameters are used to mitigate part of the modeling errors such that the specific retrieved variables of interest agree with the corresponding in situ measurements. Estimating the time-invariant variables solely from the brightness temperature observations provides a great opportunity to improve the retrievals of soil and vegetation variables on a global scale. However, the errors in the forward modeling become an evident challenge in this approach, as it could be that the surface roughness and scattering albedo values that provide the best fit in terms of brightness temperature might not produce the measured soil moisture or permittivity or vegetation optical depth. However, despite these potential issues, we found good agreement between the retrieved and measured ground permittivity and vegetation optical depth. This shows that the retrieved time-invariant surface roughness and scattering albedo are consistent with the available in situ measurements.

In the usual approach, where retrievals are performed one-by-one basis from the satellite overpasses, different errors in the observations and retrievals can be postprocessed and removed after the retrievals. In the time series approach, where the retrievals are interconnected, postprocessing is not possible, and careful preprocessing of the satellite measurement data is necessary. In our example case, the most evident issues were radio frequency interference and wet snow contamination (geophysical noise). The former contamination was removed from the data with appropriate criteria, but the latter was not, and the effects are seen in the results: on one hand, we observed errors in the retrieved vegetation optical depth during the strongest melt events, but on the other hand, the effects of the contaminated data to the time-invariant variables are not known. We observed (not shown) that without proper removal of the radio frequency interference contaminated observations the quality of the results and the convergence of the iterative algorithm to the solution were considerably degraded.

VI. Conclusion

We presented a method for retrieving a time series of geophysical variables from a time series of satellite measurements, formulated within a Bayesian framework. The presented approach has an intrinsic advantage over the usual one-by-one retrieval approach: a priori information about the temporal

characteristics of the geophysical variables can be modeled in the retrieval algorithm. In particular, the presented method allows the estimation of time-invariant geophysical variables that would not be retrieved in the one-by-one retrieval due to the lack of information in the measurements of a single satellite overpass. We demonstrated that applying the method to retrieve ground and vegetation variables from SMOS measurements yielded results consistent with both in situ measured data and with values used in previous studies. In addition to ground permittivity and vegetation optical depth, we were able to retrieve two time-invariant variables; ground surface roughness and vegetation single scattering albedo, which are often not retrieved due to the lack of information in data from a single satellite overpass. The method appeared to be sensitive to contamination of the measurements, in particular, to radio frequency interference in the SMOS data. Both the computational subjects and the approaches to account for contaminated data are potential topics for future studies.

APPENDIX

The derivative matrix of the scaled residual function F is required by the Levenberg–Marquardt algorithm, and to compute the gradient of the cost function

$$\text{cost}(w) = \frac{1}{2}\|F(w)\|^2. \tag{27}$$

Direct finite difference evaluation can be computationally infeasible. Since, in this article, we assumed $F(w) = [F_y(\varphi(w)), w]^\top$, we are interested in efficiently computing the derivative of the observation part of the scaled residual function F_y. In what follows, we will write $F_y(w) = F_y(\varphi(w))$ and $x = \varphi(w)$. To efficiently compute $\mathrm{d}F_y/\mathrm{d}w$, start with the chain rule

$$\frac{\mathrm{d}F_y}{\mathrm{d}w} = \frac{\mathrm{d}F_y}{\mathrm{d}x}\frac{\mathrm{d}\varphi}{\mathrm{d}w}. \tag{28}$$

Fix $k \in \{1, \ldots, K\}$ and denote $F_k : X \longrightarrow R^{m_k}$ the components of F_y corresponding to the satellite overpass at the time t_k. Then, using the fact that the changes in geophysical variables at a specific time t_i only affect the modeled value at the same time instance, we have

$$\frac{\mathrm{d}F_k}{\mathrm{d}x_i} = 0, \quad \text{for } i \neq k. \tag{29}$$

Here, $x_i \in \mathbb{R}^n$ is a vector representing part of the geophysical time series corresponding to the time t_i. Due to this simple notion, only the submatrices $\mathrm{d}F_k/\mathrm{d}x_k$ have to be computed in order to get $\mathrm{d}F_y/\mathrm{d}x$. The submatrices are computed by finite difference approximation and require $m_k(n+1)$ forward function evaluations each. Therefore, $\mathrm{d}F_y/\mathrm{d}x$ requires $\sum m_k(n+1) = M(n+1)$ forward function evaluations, where M is the total number of observations in the time series. The derivative of the transformation function $\varphi(w) = \exp(Aw+m)$ used in this article has a closed form expression

$$\frac{\mathrm{d}\varphi}{\mathrm{d}w} = \text{diag}(\exp(Aw+m))A \tag{30}$$

which is used to compute the final result (28).

The Laplace approximation of the posterior covariance matrix requires computing the Hessian of the cost function with respect to w. As before, direct evaluation may be computationally prohibitive. Since

$$\text{cost}(w) = \frac{1}{2}\|F_y(w)\|^2 + \frac{1}{2}\|w\|^2 \tag{31}$$

the Hessian is given by

$$\frac{\mathrm{d}^2\text{cost}}{\mathrm{d}w^2} = \frac{\mathrm{d}^2L}{\mathrm{d}w^2} + I \tag{32}$$

where $L(w) = (1/2)\|F_y(w)\|^2$ is the loss function. To compute the loss functions L Hessian in an efficient manner, start with the chain rule

$$\frac{\partial^2 L}{\partial w_i \partial w_j} = J\frac{\partial^2\varphi}{\partial w_i \partial w_j} + \left(\frac{\partial\varphi}{\partial w_i}\right)^\top H \frac{\partial\varphi}{\partial w_j}. \tag{33}$$

For the first term, the derivative of the loss function with respect to x is given by

$$J = \frac{\mathrm{d}L}{\mathrm{d}x} = F_y^\top \frac{\mathrm{d}F_y}{\mathrm{d}x} \tag{34}$$

which is computed efficiently following the previous discussion. For the second term in (33), the Hessian matrix H of the loss function with respect to x consists of $n \times n$ submatrices

$$H^{(i,j)} = \frac{\mathrm{d}^2 L}{\mathrm{d}x_i \mathrm{d}x_j}. \tag{35}$$

To compute the submatrices $H^{(i,j)}$ efficiently, write

$$\begin{aligned}\frac{\mathrm{d}^2 L}{\mathrm{d}x_i \mathrm{d}x_j} &= \left(\frac{\mathrm{d}F_y}{\mathrm{d}x_i}\right)^\top \left(\frac{\mathrm{d}F_y}{\mathrm{d}x_j}\right) + \sum_\ell F_\ell \frac{\mathrm{d}^2 F_\ell}{\mathrm{d}x_i \mathrm{d}x_j} \\ &= \left(\frac{\mathrm{d}F_y}{\mathrm{d}x_i}\right)^\top \left(\frac{\mathrm{d}F_y}{\mathrm{d}x_j}\right) + B^{(i,j)}\end{aligned} \tag{36}$$

where the summation is over the components $F_\ell : X \longrightarrow \mathbb{R}$ of F_y. Therefore,

$$H = \left(\frac{\mathrm{d}F}{\mathrm{d}x}\right)^\top \left(\frac{\mathrm{d}F}{\mathrm{d}x}\right) + B \tag{37}$$

where $\mathrm{d}F/\mathrm{d}x$ is again computed following the previous discussion. The matrix B is computed efficiently by similar reasoning: the second-order derivative matrices of F_ℓ in

$$B^{(i,j)} = \sum_\ell F_\ell \frac{\mathrm{d}^2 F_\ell}{\mathrm{d}x_i \mathrm{d}x_j} \in \mathbb{R}^{n\times n} \tag{38}$$

can be nonzero only when $i = j$. Therefore, only $B^{(k,k)}$ have to be computed for $k = 1, \ldots, K$. The $n \times n$ second-order derivative matrices of F_ℓ in the summation are computed with finite difference approximation. The sum consists of only m_k nonzero terms; those observations that correspond to the time t_k.

Finally, the first and second order derivatives of the transformation function $\varphi(w) = \exp(Aw+m)$ needed in (33) have closed form expressions given by (30) and

$$\frac{\partial^2 \varphi_k}{\partial w_i \partial w_j} = A_{k,i}A_{k,j}\exp\left(A^{(k)}w + m_k\right) \tag{39}$$

where $A^{(k)}$ is the kth row vector of the matrix A. Inserting these ingredients to (33) result in a matrix form

$$\frac{\mathrm{d}^2 L}{\mathrm{d}w^2} = A^\top D A + \left(\frac{\mathrm{d}\varphi}{\mathrm{d}w}\right)^\top H \left(\frac{\mathrm{d}\varphi}{\mathrm{d}w}\right) \tag{40}$$

where $D = \mathrm{diag}(\varphi_1 J_1, \ldots, \varphi_n J_n)$. Inserting this to (32) gives the Hessian of the cost function with respect to w.

An alternative approach to compute the Hessian is to approximate the forward model $f(\cdot)$ by linearizing it at the minimum of cost$(\cdot)$, $w = w_{\min}$. This results in the following formula (see [56]):

$$S_{w|y} = \left(S_w^{-1} + K^\top S_y^{-1} K\right)^{-1} \tag{41}$$

where $K = \mathrm{d}f/\mathrm{d}w$ at $w = w_{\min}$ and $S_w = I$ by assumption.

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